

To what extent can behavioural, spatial, temporal, and environmental features predict whether a squirrel is observed eating, and which factors are most influential in this prediction?

Group W18G03

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Executive Summary

This report analyses the 2018 Central Park squirrel census to examine whether behavioural, spatial, temporal, and environmental features can predict whether a squirrel is observed eating. The squirrel and hectare datasets were merged, cleaned, encoded, and analysed using correlation analysis, supervised learning and clustering

Overall, the results show that eating can be predicted to some extent, but not strongly. Correlation, MI and NMI found only weak relationships between individual features and eating, with movement and spatial features appearing most often. Supervised learning supported this pattern. The majority-class baseline showed that accuracy alone was misleading, because predicting all squirrels as ‘Not Eating’ found no ‘Eating’ cases. KNN had relatively high accuracy but missed many eating observations, while the Decision Tree was more useful because it identified more actual eating cases and showed that both behaviour and location contributed to prediction. Clustering found meaningful behavioural profiles, but only partly explained eating. More direct data, such as food availability, human feeding and visitor density, would likely improve prediction.

1. Introduction

This report analyses the data collected from observers, citizens, over a period of two weeks in 2018 about Eastern Gray squirrels from New York City’s Central Park. The squirrel dataset contains information about where a squirrel was spotted, physical and behavioural characteristics of the squirrel observed. The hectare dataset contains information about the area in which the squirrel was observed such as ‘Hectare Conditions’, whether it was busy or calm, ‘Total Time of Sighting’. The purpose of the report is to investigate the extent to which behavioural, spatial, temporal, and environmental features can be used to predict whether a squirrel is likely to be observed eating. In the analysis, the data first underwent preprocessing to transform the two of the raw datasets provided into a single, clean, and numerically encoded dataset. Following that, to study the relationships behavioural features and location had with eating, correlation analysis was conducted. Furthermore, supervised learning models were implemented to predict whether a squirrel was eating. Clustering the squirrels based on behavioural, and temporal and environmental features was conducted to verify the results obtained.

2. Methodology

2.1 Data Preprocessing

The two raw datasets provided (squirrel.csv, hectare.csv) were merged into a single, clean, and numerically encoded dataset suitable for further. A left join on the composite key (Hectare, Shift, and Date) was used to keep every squirrel observation and match it with hectare context where possible. The composite key prevents incorrect many-to-many matches across sessions in the same hectare. Nine

squirrel rows had no matching hectare session and were imputed in the next steps. From hectare.csv only ‘Hectare Conditions’, ‘Number of Squirrels’, ‘Total Time of Sighting’, and ‘Number of sighters’ were retained. The remaining columns were removed (Table 2.1.1).

Table 2.1.1: Columns excluded from hectare.csv and their respective justification for removal

Column/Feature	Justification for Removal
Litter	Has ~54% missing, too sparse to impute reliably
Litter Notes/ Hectare Conditions Notes	Has >90% missing
Other Animal Sightings	Mixed test
Sighter Observed Weather Data	Inconsistent entries (e.g. ‘sunny’, ‘Sunny’, ‘clear’ / use of Fahrenheit and Celsius); minimal weather variation across two weeks of the observation
Anonymized Sighter	Observer’s identifier, no relation to squirrel eating behaviour

After merging, columns irrelevant to the research question were dropped: identifiers (no predictive value) and columns with over 80% missing values were dropped (unreliable imputation, introduce noise). Table 2.1.2 lists the removals.

Table 2.1.2: Columns excluded from the merged dataset and their respective justification for removal

Column/Feature	Justification for Removal
Unique Squirrel ID/ Hectare Squirrel Number	Identifier column, carry no predictive information
Highlight Fur Color	High cardinality
Combination of Primary and Highlight Color	Combination of Primary and Highlight Fur Color
Color notes/ Specific Location/ Other Activities/ Other Interactions	Has >80% missing values
Lat/Long	Redundant, already captured by X and Y columns

Feature selection was guided by the research question. Behavioural variables were retained as they describe activities that may occur with or instead of eating. Spatial variables for capturing differences in food availability, vegetation, or observation context. Temporal and hectare-level variables as contextual predictors. Features were removed only when they were identifiers, highly incomplete, redundant, or too unstructured to encode reliably.

Handling Missing Values

Each imputation was based on the missingness rate and the data type of the column.

‘Age’ contained 121 missing values (4.00%) plus 4 entries recorded as “?”. The “?” entries were recorded to a separate “Unknown” category since the observer uncertainty is different from absent data. The nulls were imputed with the mode. Random sample imputation was considered as it would preserve the original distribution, but with only 4.00% missing, the two approaches are practically equivalent. ‘Primary Fur Color’ (55 missing, 1.82%) and ‘Location’ (64, 2.12%) were similarly imputed with their respective modes. At these small proportions, row deletion would be wasteful. ‘Hectare Conditions’ contained two similar categories describing transitional activity levels: ‘Calm, Busy’ and ‘Medium’. Therefore, both were merged into ‘Moderate’ prior to imputation. The remaining missing values, (163 rows, 5.39%), were imputed with the mode.

‘Number of Squirrels’ (9 missing, 0.29%) and ‘Number of sighters’ (9 missing, 0.29%), and ‘Total Time of Sighting’ (76 missing, 2.51%) were imputed with their respective medians. Median was preferred over mean as it is robust to outliers (long sighting sessions would skew the mean upward and produce an unrepresentative value).

Feature Engineering

Two features were engineered. The ‘Above Ground Sighter Measurement’ consisted of a mix of “FALSE” strings (ground-level observations) and numeric strings. A new ‘Above Ground Height’

column was added by converting numeric to float and replacing “FALSE” with 0.0 to preserve spatial information and enable numerical analysis. The ‘Date’ column was stored as integer in MMDDYYYY format. It was parsed to extract a binary Is Weekend (1=weekend, 0=weekday). Weekends bring higher human foot traffic to Central Park, which may affect squirrel and human interaction worth analysing. Extracting day-of-week was rejected as duration of 14 days give too few samples per weekday to be useful.

Outlier Inspection

Continuous features were inspected for outliers using boxplots and the IQR rule. All flagged values were retained as they represent valid observations rather than outliers (e.g. dense hectares, long observation sessions and tall sightings). Removal would discard possible insight into research question.

Encoding and Scaling

Several columns (e.g. ‘Runing’, ‘Climbing’ etc) recorded Boolean values representing presence or absence of specific behaviours. These were converted to integers (0=absence, 1=presence) for compatibility with learning models. One-hot encoding was not considered as the columns were already binary and one-hot encoding would add redundant columns. Binary categorical column ‘Shift’ (AM/PM) and ‘Location’ (Ground Plane/Above Ground) were label encoded as 0 and, since no order is implied. Columns with more than two categories were one-hot encoded to avoid implying an order between the categories. For instance, label encoding Gray=0, Cinnamon=1, Black=2, would suggest false orders (Black > Cinnamon > Gray). Standardisation was applied to ensure fair comparison for distance-based models such as KNN and K-means. Without scaling, features with larger numerical scale (X, Y, or time), would dominate the distance calculations and may bias the predictive behaviour and clustering outcomes. Min-max normalisation was rejected as standardisation is more robust to outliers. For the supervised learning models, scaling was applied on the training set only and applied to validation and test sets to prevent data leakage. This ensures that the model never accesses the information from unseen data during training, which would otherwise produce wrong estimates.

2.2 Correlation

Correlation analysis was used prior to modelling to identify which features showed any signal with Eating. Features were grouped based on the research question so that both individual and group-level patterns could be compared. Three methods were used, since each capture a different aspect of association as summarized in Table 2.2.1.

Table 2.2.1: Summary table of different correlation method

Method	Main purpose	Suitable data	Result	Limitation
Pearson (r)	Linear association	Continuous	Direction and strength [-1,1]	Not ideal for discrete variables; only check for linear relationship
Mutual Information (MI)	Shared information	Discrete or discretized	Magnitude of dependency	No direction: raw score can be hard to compare
Normalized Mutual Information (NMI)	Normalized MI	Discrete or discretized	Shared information on [0,1]	No direction

Pearson correlation

Pearson correlation was used as an initial check of direction and linear strength. Since most features are binary or one-hot encoded after preprocessing, Pearson is not ideal as a final measure (reduces to phi coefficient for binary vs. binary and point-biserial for binary vs. continuous). The values should only be read as indicators of co-occurrence direction rather than linear association. Spearman was considered but rejected because most of the variables are not ordinal after preprocessing, so rank-based correlation adds no information.

K-bins

Before calculating MI and NMI, the continuous variables ‘X’, and ‘Y’ were discretized with three bins and quantile (equal frequency) strategy. Three bins balance interpretability (low, medium, high) with stable bin counts. Quantile was preferred over equal-width because ‘X’ and ‘Y’ are non-uniformly distributed across the park and equal-width binning would have left some bins almost empty and would make the MI scores less stable. ‘Number of Squirrels’ and ‘Number of sighters’ were count variables, so they were already discrete. For variable ‘Above Ground Height’ also had many repeated 0 values, which made k-bins discretization less stable.

Mutual Information and Normalized Mutual Information

MI was used as a complementary measure because it captures non-linear dependencies. This was more suitable for the processed dataset, where many variables were binary or one-hot encoded. MI was calculated for both individual features and feature groups. For the group level comparison, the average MI score was used rather than the total score because the feature groups contained different number of variables. This made the comparison fairer and avoided giving larger groups an advantage because they had more features. NMI was used after MI to make the feature scores easier to compare across variables. Raw MI scores can be affected by the entropy or variation of each feature, so features with more possible values may naturally produce larger MI scores. NMI reduced this issue by normalizing the scores, making the comparison across different types of features more balanced.

Feature-feature heatmaps

Pearson and NMI heatmaps were generated for a subset of interpretable predictors covering main feature types (behaviour, location, height, shift, and human interaction). These heatmaps were not used to remove features directly, but to check whether predictors carried overlapping information and to support later interpretation of feature relevance and model result. A full heatmap across all predictors was rejected given the high number of features after encoding.

2.3 Supervised Learning

Two supervised models were trained to predict whether a squirrel is observed eating: a Decision Tree (DT) and K-Nearest Neighbours (KNN). These two were chosen because they approach classification differently; KNN makes distance-based predictions while DT is model-based and interpretable. If they agree on which features matter, that agreement is more credible, and if they disagree, then the disagreement itself is informative about the structure of the data. Random Forest was considered but not implemented. Although it can reduce the risk of overfitting, its structure makes it harder to inspect specific decisions. On the other hand, a DT provides branching and thresholds which are more useful in explaining the influence of factors.

Experimental Design and Evaluation Strategy

The task was treated as a binary classification problem with target variable ‘Eating’, which was removed from the predictors. The remaining processed variables were used as input features. The dataset was split into training (70%), validation (10%), and test (20%) sets using stratified sampling to preserve 25.1% eating class ratio in all three sets. The training set was fit the models; the validation set tuned hyperparameters, and the test set was kept separate for final evaluation.

Table 2.3.1: Supervised learning model setup

Model	Reason for use	Parameter tuned	Scaling
Decision Tree	Interpretable rule-based model, useful for feature importance	max_depth	Not required
KNN	Distance based comparison model	k / n_neighbors	Required

For both models, optimal hyperparameters were selected using validation F1-score rather than accuracy. Because eating is the minority class, a model that predicts ‘not eating’ would achieve 74.9% accuracy while identifying no ‘Eating’ squirrels. F1-score balances precision, which measures how many

predicted eating observations were eating, and recall, which measures how many actual eating observations were identified. A majority-class baseline that always predicts 'Not Eating' was included as a reference point to verify that DT and KNN learned useful eating patterns beyond the class imbalance. A held-out validation set approach was selected over k-fold cross-validation for hyperparameter tuning. While cross-validation provides more stable estimates by averaging across multiple folds, it introduces additional complexity in ensuring that preprocessing steps such as scaling are applied correctly within each fold to avoid leakage. Given the dataset size of 3023 observations, a single validation set produces reliable estimates without added risk.

k-Nearest Neighbours and Decision Tree

KNN predicts each observation based on its nearest neighbours in the training data. Because KNN is distance-based, the input features were standardised before fitting. The optimal k was found by evaluating F1-score on the validation set across odd values from 3 to 31 (odd values to avoid ties). k=1 was excluded as each training point's nearest neighbour is itself, so it yields near perfect performance but performs poorly on unseen data. A Decision Tree was selected because it can separate eating and non-eating observations using decision rules and show which features were important in making the splits. The `max_depth` parameter was tuned across 1-19. Entropy was used as the splitting criterion, and `class_weight='balanced'` was applied to recover minority cases.

Feature Importance

Feature importance was examined after training, using built-in feature importance for the DT and permutation importance for KNN. Cross-validation was considered for tuning, but the train-validation-test split was kept because it provided a clear validation set for parameter tuning and a separate test set for final evaluation.

2.4 Clustering

The first cluster analysis was done on behavioural features ('Running', 'Chasing', 'Climbing', 'Foraging', 'Tail Flags', 'Tail Twitches', 'Approaches' (humans), 'Indifferent', 'Runs From' (humans) and 'Sounds'). 'Eating' was excluded as the goal was to identify behavioural patterns associated with eating rather than clustering on eating. The three behaviours 'Kuks', 'Quaas' and 'Moans' were condensed into one column, 'Sounds'. If the squirrel was observed making any of those three noises, 'Sounds' had a value of 1, otherwise 0. This was done to reduce the dimension of the data to allow for better grouping of clusters since such a small percentage of squirrels displayed those features. The second clustering analysis was done on temporal features, 'Shift' and 'Is Weekend' and some spatial features, 'Hectare Conditions_Calm', 'Hectare Conditions_Moderate', 'Hectare Conditions_Busy'. All the selected features are binary and variables that were standardised were excluded so that all the clustering inputs had the same format, binary values, and hence are treated with equal weight.

Both analyses used K-Means clustering which groups similar data points based on their Euclidean distance to the centroid. Hierarchical clustering was considered as an alternative approach. Hierarchical clustering produces a dendrogram to visualise how clusters merge and relationships between different clusters. Although it is less sensitive to outliers compared to K-Means, it is significantly more computationally expensive. Hence K-Means was used, as this is a large data set.

The Elbow Method was used to determine the optimal number of clusters as it clearly shows how the tightness of clusters changes as the number of clusters changes. K-Means was run from 1 to 10 clusters, with a fixed seed for reproducibility. The distance from each point in the cluster to its centroid was calculated, squared then summed to provide a distortion figure. The distance was squared so that positive and negative distances do not cancel each other out and so that larger errors are penalised more heavily. The distortion is plotted against the number of clusters, and the point where increases the number of clusters does not give meaningful improvement in the distortion was chosen as the optimal number of clusters.

For the first clustering analysis, to explore the characteristics for each cluster, the top 3 most likely and unlikely features were plotted using bar charts where the x-axis is the cluster number and the y axis is the proportion of squirrels in that cluster exhibiting a certain behaviour. For the second clustering analysis, due to the small number of features the proportion of figures present in each cluster was ranked.

Per cluster, the percentage of squirrels observed eating was calculated to determine whether certain groups of behavioural features were associated with higher or lower rates of eating. A scatterplot was produced to visualise the distribution of clusters, grouped on behaviour, across the X and Y coordinates. This allowed for visual analysis of identifying any location-based groupings.

3. Results Exploration and Analysis

3.1 Data Preprocessing

Class Distribution and Final Dataset

Following all the preprocessing steps, the merged and cleaned dataset contains 3023 rows and 31 columns. The final features can be divided into five main categories summarised in Table 3.1.1.

Table 3.1.1: Final feature set after preprocessing, grouped by category

Category	Count	Features
Behavioural	12	Running, Chasing, Climbing, Foraging, Kuks, Quaas, Moans, Tail flags, Tail twitches, Approaches, Indifferent, Runs from
Spatial	4	X, Y, Location, Above Ground Height
Temporal	2	Shift, Is Weekend
Environmental	7	Number of Squirrels, Number of sighters, Total Time of Sighting, Hectare Conditions_Calm, Hectare Conditions_Moderate, Hectare Conditions_Busy
Demographic	6	Age_Juvenile, Age_Adult, Age_Unknown, Primary Fur Color_Black, Primary Fur Color_Cinnamon,
Target	1	Eating

The target variable distribution shows a moderate class imbalance 3:1 which informs every subsequent modelling choice (stratified splitting, balanced class weight, F1 selection).

Table 3.1.2: Imbalanced distribution of Eating across observations.

Eating	Proportion
Not Eating	74.9%
Eating	25.1%

3.2 Correlation

Figure 3.2.1 shows the top 10 features ranked by Pearson correlation with Eating. All values were weak: the strongest associations were negative, with 'Running', 'Climbing', and 'Above Ground Height'. Positive associations were even weaker, with 'Approaches' and 'Indifferent' both below $r = 0.07$. As a benchmark, $|r| < 0.15$ is generally considered a very weak association and could reflect sampling variation rather than a stable underlying relationship. No single feature shows a strong linear signal with Eating; the dominant pattern is simply that movement-related and above-ground observations tend not to be recorded together with eating.

Figure 3.2.2 compares the average absolute Pearson correlation by feature group. Spatial features had the highest average, followed by behavioural and temporal; environmental and characteristic features were the weakest. This suggests that location and movement carry most of the weak signal, but the absolute magnitude remain small so it should be read as relative ranking only.

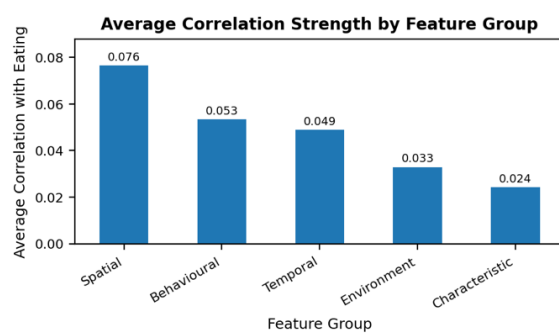
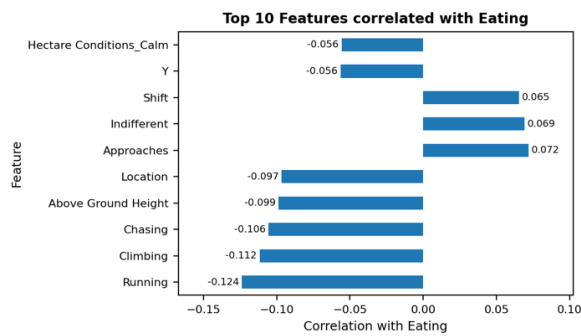


Figure 3.2.1: Top features by Pearson correlated with eating **Figure 3.2.2:** Average Pearson correlation by feature

Figure 3.2.3 shows the top 10 features ranked by MI score. The highest-ranked features were ‘Above Ground Height’ followed by ‘Total Time of Sighting’, ‘Running’, and ‘Climbing’. The MI ranking confirms the Pearson pattern (movement and spatial features dominate). However, the MI scores were very small, which suggests that no single feature shared a large amount of information with ‘Eating’.

Figure 3.2.4 compared the average MI score for each feature group. The spatial group had the highest average MI score, followed by environmental and behavioural groups. This indicated that location-related features, observation context, and behaviour carried relatively more information than the other feature groups. However, the MI scores were still very small. Therefore, these results did not show a strong relationship between any single feature or feature group and ‘Eating’. Instead, they suggested that the useful information was spread across several weak predictors, which supported using these variables together in the later supervised learning models.

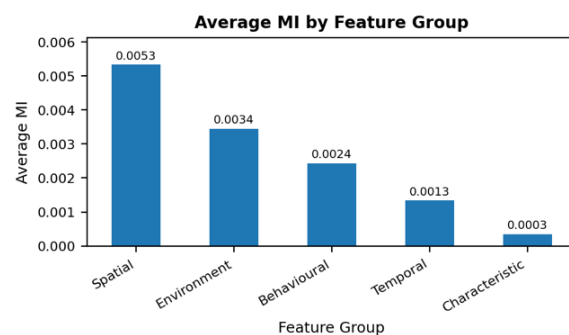
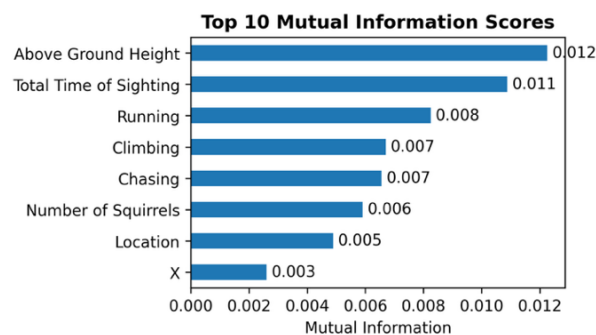


Figure 3.2.3: Top 10 features ranked by Mutual Information **Figure 3.2.4:** Average MI score by feature group

Normalized Mutual Information was used after MI to make the feature comparison more balanced across variables. Figure 3.2.5 showed that the highest NMI scores came mainly from movement-related and spatial features, including ‘Chasing’, ‘Running’, ‘Climbing’, ‘Location’, and ‘Above Ground Height’. However, the highest score was only around 0.015, so these features were still weak signals rather than strong predictors.

Figure 3.2.6 compares the average NMI score by feature group. The spatial group had the highest NMI score, followed by the behavioural and environmental groups. This indicated that location-related features and movement-related features behaviours may provide relatively more information about whether eating was recorded. However, the group scores were also small, so this did not mean that any one group could explain eating behaviour on its own. Overall, NMI supported the same conclusion as Pearson and MI. No single feature or feature group strongly explained ‘Eating’. This meant these variables may be useful in the models, but mainly when combined with other weak signals.

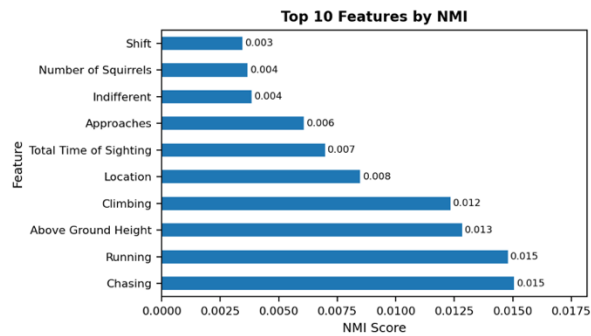


Figure 3.2.5: Top 10 features by NMI

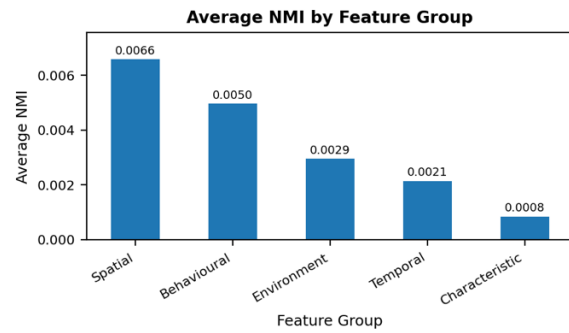


Figure 3.2.6: Average NMI score by feature group

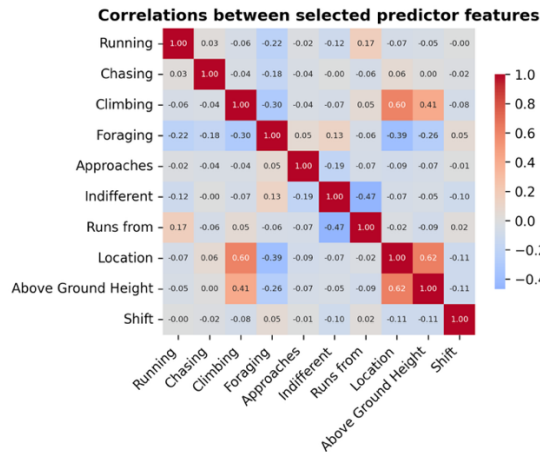


Figure 3.2.7: Pairwise Pearson correlation

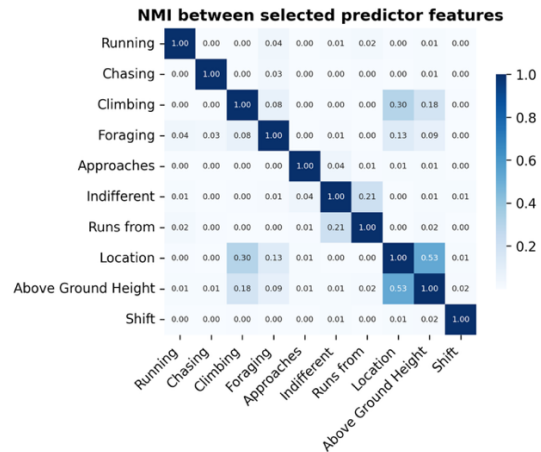


Figure 3.2.8: Pairwise Normalized Mutual Information

Figure 3.2.7 (Pearson) and Figure 3.2.8 (NMI) show pairwise relationships among selected predictors. The strongest overlaps were between ‘Location’ and ‘Above Ground Height’, and between ‘Climbing’ and ‘Location’. This is intuitive: above-ground observations are largely climbing observations on trees. Foraging was negatively correlated with all three. This suggests that foraging was more often recorded when squirrels were closer to the ground. These overlaps matter for interpreting later model results. Because ‘Location’, ‘Climbing’, and ‘Above Ground Height’ carry redundant information, a distance-based model such as KNN effectively counts the same signal three times in its distance calculation.

3.3 Modelling

Hyperparameter Tuning

DT validation F1-score (Figure 3.3.1) increased with depth, maximising at max_depth=15. KNN validation F1 (Figure 3.3.2) fell with k and was maximised at k=3.

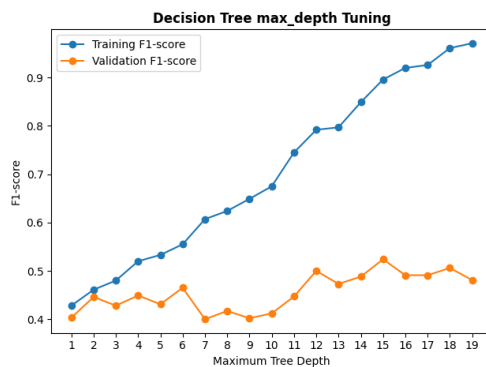


Figure 3.3.1: Decision Tree max_depth tuning.

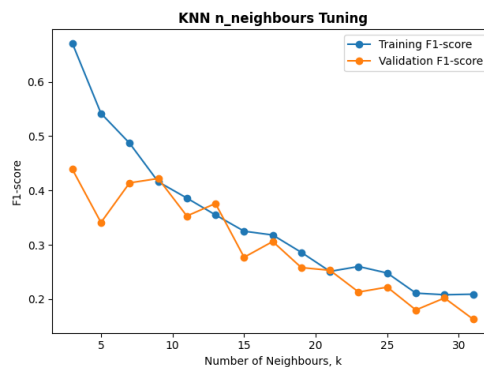


Figure 3.3.2: KNN n_neighbors tuning.

Test Set Performance

Test-set performance was evaluated using accuracy, precision, recall and F1-score. Recall and F1-score were prioritised as the goal was to find actual Eating observations, not just achieve high accuracy by predicting the majority ‘Not Eating’ class. Figure 3.3.3 shows results baseline, DT and KNN. The baseline achieved the highest accuracy by always predicting ‘Not Eating’, but scored 0 on precision, recall, and F1-score for Eating class. This shows that accuracy alone is misleading on imbalanced data. KNN had similarly high accuracy, but low recall for Eating. The DT had slightly lower accuracy, but much higher recall and F1-score. This means that DT is more effective for identifying eating squirrels.

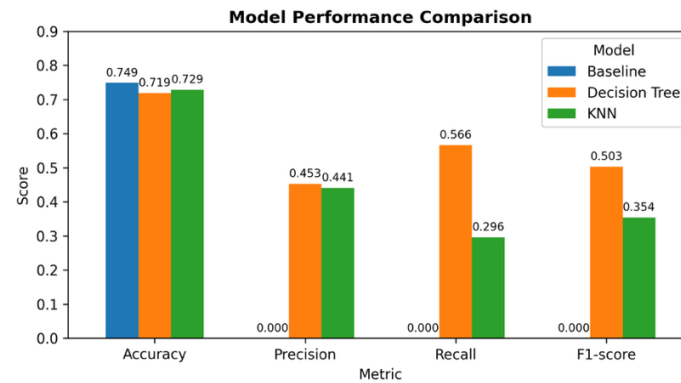


Figure 3.3.3: Test-set performance comparison

Figure 3.3.4 shows the confusion matrices for the baseline, DT and KNN. The baseline predicted every observation as ‘Not Eating’, so it missed all actual ‘Eating’ cases despite high accuracy. KNN correctly predicted 396 ‘Not Eating’ observations, compared with 349 for the DT. However, KNN only correctly identified 45 ‘Eating’ observations and missed 107 actual ‘Eating’ cases. In comparison, the DT correctly identified 86 ‘Eating’ observations and missed 66 ‘Eating’ cases. Overall, KNN was stronger on the majority class, while the DT was more useful for detecting actual Eating observations.

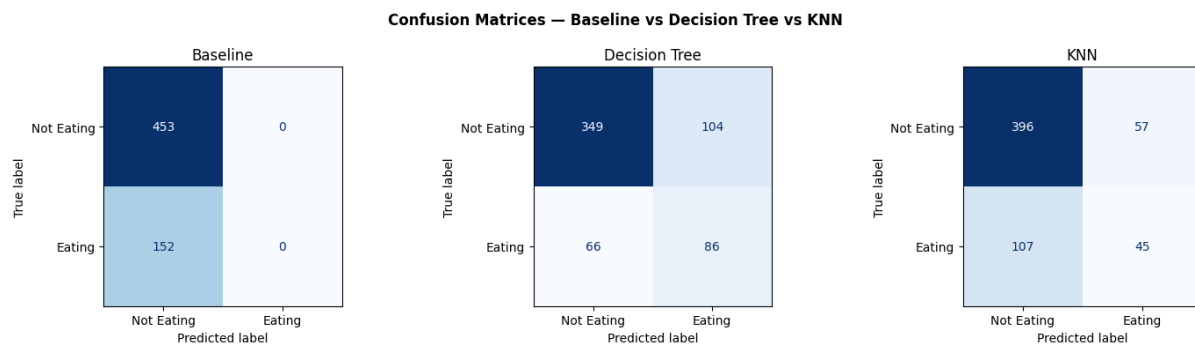


Figure 3.3.4: Test-set confusion matrices.

Feature Influence

The two models diverged in which feature importance (Figure 3.3.5). The DT ranked spatial coordinates (X, Y) highest, followed by environmental features (Number of Squirrels, Total Time of Sighting), and behavioural features (Foraging and Chasing). This suggests that DT combined location and observed behaviour for predictions. KNN permutation importance ranked ‘Running’, ‘Foraging’, and ‘Is Weekend’ at the top. High permutation importance for ‘Running’ and ‘Foraging’ means that shuffling these variables reduced the model’s F1-score the most. Therefore, behavioural features were important for the distance-based KNN model.

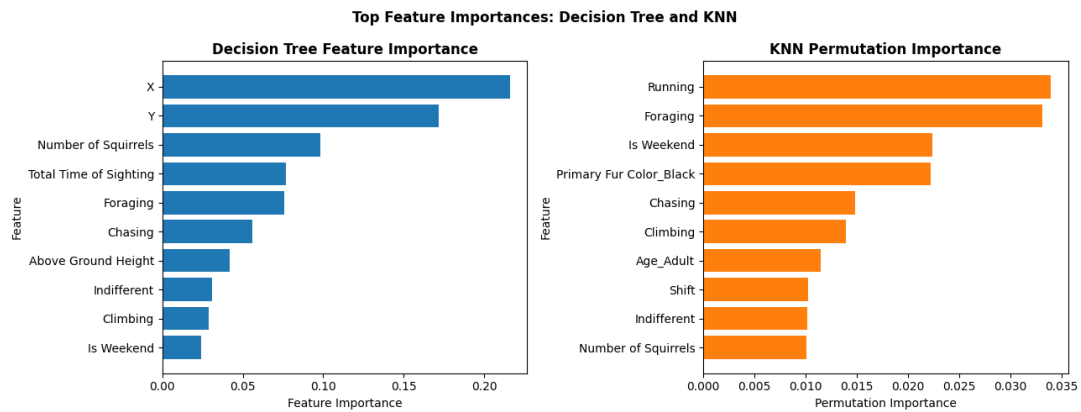


Figure 3.3.5: Top 10 feature importance for the Decision Tree (left) and KNN

The structure of the tree (decisions, nodes, branches, and thresholds) is shown in Figure 3.3.6. The root node splits on ‘Running’ where it divides 2418 training samples into two groups. This means ‘Running’ was the first rule used by the tree to separate the observations, although it did not have the highest overall feature importance. This is consistent with the earlier correlation finding that ‘Running’ had the strongest negative association with ‘Eating’.

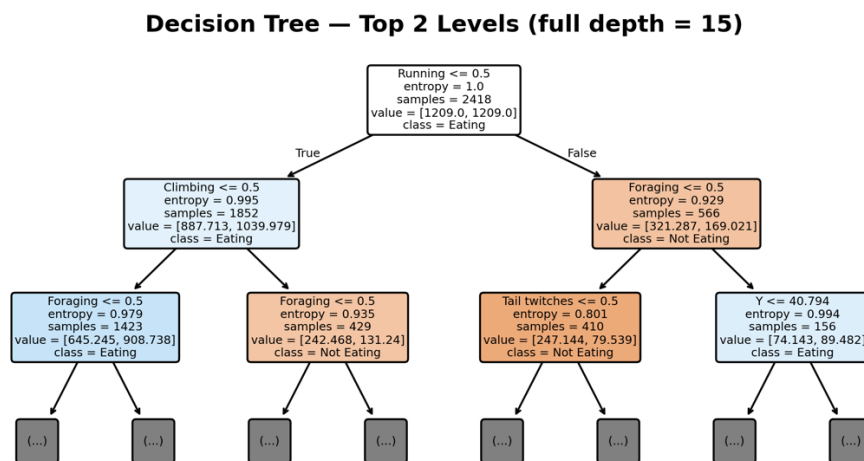


Figure 3.3.6: First three levels of the tuned Decision Tree (max_depth=15 in full, depth=3 shown for readability).

3.4 Clustering

Clustering on behavioural features

From Figure 3.4.1, the optimal number of clusters is 4 as the decrease in distortions fairly small as the number of clusters increases from 4 to 5 compared to previous decreases in distortions as k increases.

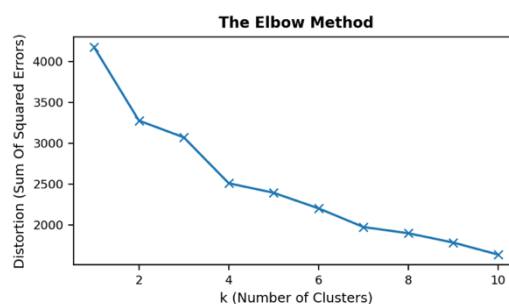


Figure 3.4.1: The Elbow Method for k-means clustering on behavioural features (k=1-10)

For each of the four clusters, the top 3 most exhibited behaviours and least exhibited behaviours were determined.



Figure 3.4.2: Behaviours observed in Cluster 0

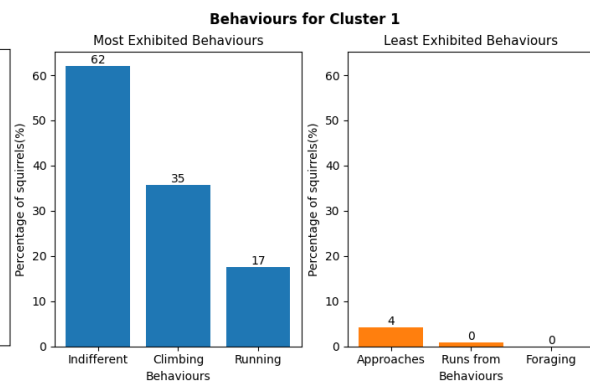


Figure 3.4.3: Behaviours observed in Cluster 1

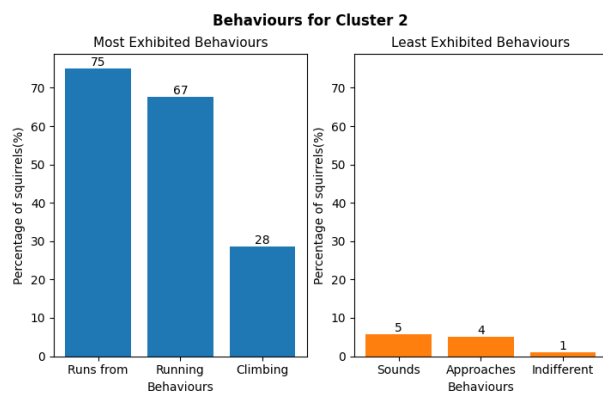


Figure 3.4.4: Behaviours observed in Cluster 2

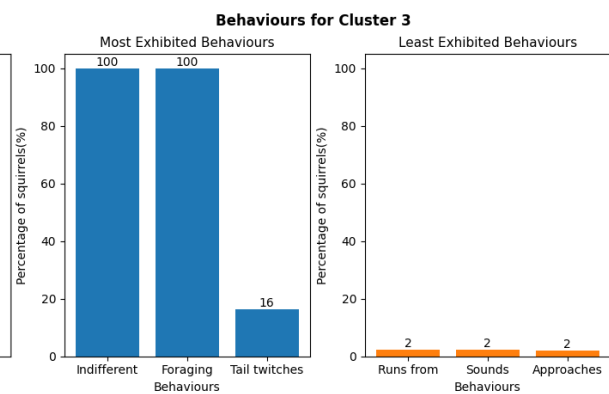


Figure 3.4.5: Behaviours observed in Cluster 3

Based on the top likely behavioural features per cluster from Figures 3.4.2 -3.4.5, each cluster represents a behavioural profile of squirrels. Cluster 0 are **foraging, human-avoidant** squirrels. This suggests that squirrels in cluster 0 are focused on seeking food or other materials but are cautious around humans. Cluster 1 are **non-foraging, human-indifferent** squirrels. This could be interpreted as squirrels in this cluster being comfortable with the presence of humans enough to carry on their activities as normal. Cluster 2 are **non-foraging, human-avoidant** squirrels. Perhaps squirrels in this cluster are more sensitive to human presence and are running away from humans rather than performing their normal activities. Cluster 3 are **foraging, human-indifferent** squirrels. This suggests that squirrels in cluster 3 are focused on seeking food or other materials and are not bothered by human presence.

Table 3.4.1: Percentages of squirrels eating in each of the different behavioural clusters

Cluster	Eating (%)	Difference from Average (%)
0	23.9	-1.2
1	28.8	+3.7
2	19.7	-5.4
3	25.2	0.1

In Table 3.4.1, the percentage of squirrels in each cluster was compared against the average of 25.1%.

The clusters were plotted against the location of each squirrel where it was observed to identify any patterns between behavioural features and location as in Figure 3.4.6. The data is in a slanted rectangular shape as the squirrels were observed within the coordinates of the park and there is a white space in the middle as a reservoir. There is no clear separation of clusters which signifies that behaviours (excluding eating) are not significantly more likely to occur at region of the park accessible to squirrels over another.

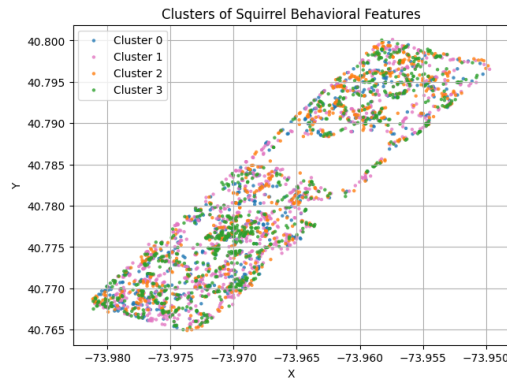


Figure 3.4.6: Distribution of behavioural clusters based on location

Clustering on temporal and some environmental features

From Figure 3.4.7, the optimal number of clusters was determined to be 4 as the decrease in distortions fairly small as the number of clusters increases from 4 to 5.

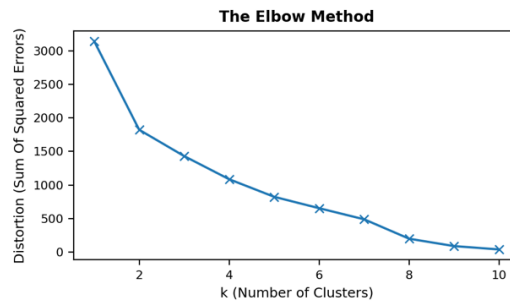


Figure 3.4.7: The Elbow Method for k-means clustering on temporal and some spatial features (k=1-10)

For each of the clusters, the percentage of a certain feature exhibited within that cluster was plotted.

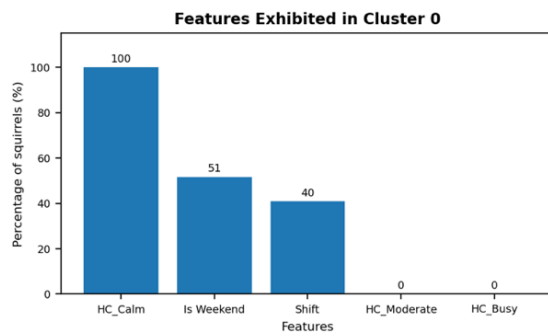


Figure 3.4.8: Features Exhibited in Cluster 0

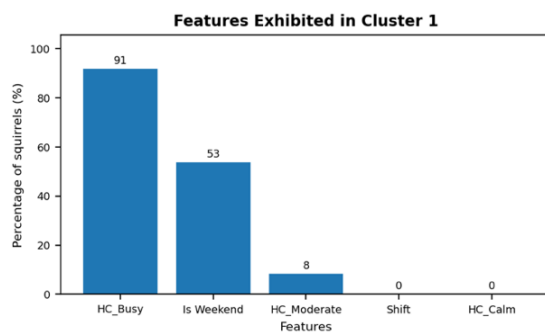


Figure 3.4.9: Features Exhibited in Cluster 1

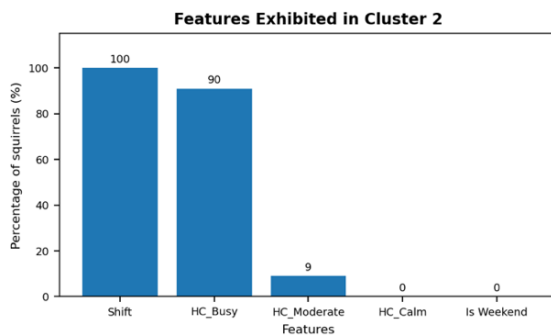


Figure 3.4.10: Features Exhibited in Cluster 2

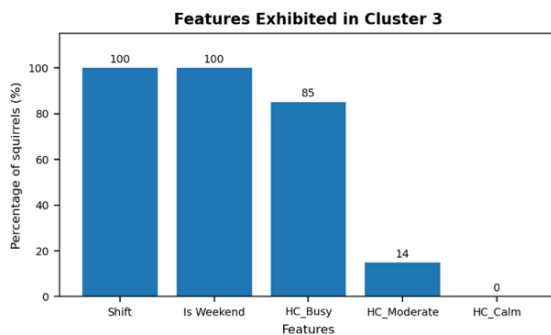


Figure 3.4.11: Features Exhibited in Cluster 3

From Figures 3.4.8 – 3.4.11, each cluster represents a different profile of temporal and environmental features. Cluster 0 have **minimal environment activity, with observations majorly on the weekend**. Observations in this cluster were made when ‘Hectare Conditions’ was “Calm” and roughly half of these observations were made on the weekend (‘Is Weekend’ is 1). Cluster 1 have a **busy environment, with observations majorly on the weekend**. Observations in this cluster were made when ‘Hectare Conditions’ was “Busy” and roughly half of these observations were made on the weekend (‘Is Weekend’ is 1). Cluster 2 have a **busy environment, with observations during PM shifts**. Observations in this cluster were made in the afternoon and evening (‘Shift’ is 1) and when ‘Hectare Conditions’ was “Busy”. Cluster 3 have a **busy environment, with observations during PM shifts and the weekend**. Observations in this cluster were made in the afternoon and evening (‘Shift’ is 1) on the weekend (‘Is Weekend’ is 1) where ‘Hectare Conditions’ was “Busy”.

Table 3.4.2: Percentages of squirrels eating in each of the different temporal and environmental clusters

Cluster	Eating(%)	Difference from Average (%)
0	22.0	-3.1
1	23.9	-1.2
2	29.4	+4.3
3	28.2	+3.1

In Table 3.4.2, the percentage of squirrels in each cluster was calculated and compared against the average of 25.1%. The clusters were plotted against the location of each squirrel to identify any patterns between temporal and environmental features and location in Figure 3.4.12. While there are a few groupings of colours in specified regions, this occurs all over the park for each cluster type hence there are no clear boundaries between clusters.

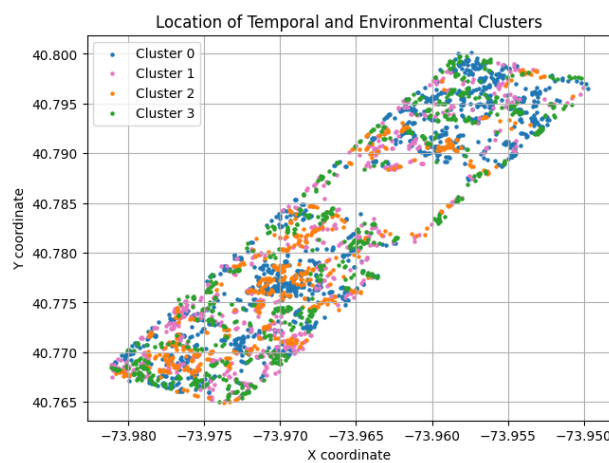


Figure 3.4.12: Distribution of temporal and environmental clusters based on location

4. Discussion and Interpretation

4.1 Preprocessing

Several preprocessing decisions shaped the results. ‘Is Weekend’ justified its inclusion by ranking third in KNN permutation importance. This suggests weekend foot traffic patterns carry temporal signal relevant to eating behaviour. Standardisation influenced KNN as ‘Running’ and ‘Foraging’ were found to be the top predictors, which would likely be dominated by ‘X’ and ‘Y’ without scaling. Merging ‘Calm, Busy’ and ‘Medium’ into ‘Moderate’, reduced ‘Hectare Condition’ from 5 categories to 3. If eating rates were different between these two conditions, merging them made them indistinguishable for correlation analysis. One-hot encoding allowed categorical features to be included across all analysis without false orderings. However, it increased the dimensionality, making KNN less effective in

differentiating near and far neighbours as distance between two points becomes similar in high dimensions.

4.2 Correlation

The correlation results showed that eating behaviour was difficult to explain using the available variables alone. The strongest signals (Running, Climbing, Chasing, Above Ground Height) are all negatively associated with Eating. This does not mean that movement prevents Eating (squirrels stop moving to eat, but they also stop moving to do many other things). A more likely explanation is that an observer tends to record one main activity per sighting so a squirrel logged as 'Running' is unlikely to also be logged as Eating in the same record. The negative correlations reflect how data was recorded rather than a relationship between movement and eating.

Several factors of Eating are not measured in the dataset. Food availability varies across the park but is not recorded directly. Human feeding interactions are absent. Observer's attention may be biased as an observer focused on a running squirrel may not notice subtler activities such as eating. These unmeasured factors limit how much correlation can contribute to analysis.

Correlation analysis identifies which features co-occur with eating but cannot say whether those features would predict eating when considered together. The weak individual signals imply that either eating is largely unpredictable from the available features, or the signal is distributed across many weak predictors that only become useful in combination. The correlation analysis on its own only partly answers the question: behavioural and spatial features are the most relevant groups, but the strength of the answer depends on whether modelling can aggregate the weak signals into reliable prediction. Overall, the correlation section only partly answered the research question.

4.3 Modelling

The learning models' results show that behavioural and spatial features carry limited predictive information. This answers the research question only partly. Because 'Eating' made up only 25.1% of the dataset, the models had far more 'Not Eating' examples to learn from. This made 'Eating' harder to predict compared with 'Not Eating'. This matches the correlation results, where individual features were only weakly related to Eating. The supervised models tested whether these weak signals could work together, rather than relying on one strong predictor.

Although KNN had slightly higher accuracy, KNN was mainly better at recognising the majority 'Not Eating' class. It missed many actual 'Eating' observations, showing that the features were not strong enough for the distance-based model to clearly separate 'Eating' from 'Not Eating'. The DT gave a more useful result; it identified more actual 'Eating' observations with better recall and F1-score for the 'Eating' class. The DT also made more false 'Eating' predictions, partly because `class_weight = 'balanced'` gave more weight to the minority 'Eating' class, improving recall but increasing false positives. Therefore, the features were helpful for detecting possible 'Eating' cases, but they were not reliable enough to perfectly identify 'Eating' behaviour.

Another possible reason is that some features related to 'Eating' are not the same as eating itself. For example, foraging is closely related to searching for food, but a squirrel that is foraging may not be eating at the exact observation moment. Similarly, certain X and Y locations may be associated with more food or human feeding, but not every squirrel in those areas will be eating.

Feature importance results suggest that the research question cannot be answered by a single feature alone. The DT relied more on spatial features, while KNN relied more on behavioural features. This suggests that 'Eating' was predicted using both where the squirrel was observed and what it was doing. Behaviour described the squirrel's immediate activity, while location gave context about where 'Eating' was more likely to be observed.

The overlapping signal between 'Location', 'Above Ground Height', and 'Climbing' identified in the correlation analysis may partially explain why two models relied on different features. Because KNN calculates the total distance across all features, the signal is counted three times in calculation. However,

with permutation importance, KNN's performance drops the most when 'Running' is shuffled as it was less redundant than the overlapping features.

Despite using held-out test, the gap between the DT's training F1 (~0.95) and validation F1 (~0.55) shown in Figure 3.3.1, indicates overfitting even with max_depth tuning. This limits generalisation on unseen data. A shallower depth could have been explored, though at the cost of identifying eating observations. Another limitation is the class imbalance. Despite applying 'balanced' class_weight and selecting F1 as evaluation metric, the best Eating F1-score was only 0.503, and the recall of 0.566 showed that many actual 'Eating' observations were still missed. Oversampling techniques such as SMOTE could have been explored as an alternative. Random Forest may have generalised better by combining multiple trees, but this would come at the cost of the interpretability that made the DT useful for this analysis.

Overall, supervised learning showed that behavioural, spatial, temporal, and environmental features can predict 'Eating' to some extent, but not strongly enough to fully answer the research question.

4.4 Clustering

Clustering on Behavioural Features

From Table 3.4.1, cluster 0 and 3 have very similar percentage of squirrels eating of 23.9% and 25.2%. The top feature both clusters have in common is 'Foraging'. However, in cluster 0, many squirrels run from humans while in cluster 3, squirrels are indifferent to human presence. If squirrels are foraging, their reaction to humans is mostly not influencing their likelihood of eating. Cluster 2 has below average 19.7% of squirrels eating. They might run away from humans rather than eating. 'Running' is also the most negatively correlated variable with 'Eating', further validating why cluster 2 has the lowest percentage. Cluster 1 has the highest percentage of squirrels eating of 28.8%. Although cluster 3 squirrels are also indifferent to humans, it does not have a much higher than average percentage eating which is unexpected as squirrels in that cluster forage, the act of searching for food (or other) resources.

Clustering on Temporal and some Environmental Features

Clusters with busy hectare conditions have higher rates of squirrels eating than calm conditions, as seen with cluster 1, 2 and 3. Squirrels in high activity areas might be more likely to eat as they feel safer eating due to reduced predator presence as humans scare predators away (Chelsea Ortiz-Jimenez, 2021). Additionally, observations in the PM shift, clusters 2 and 3, recorded higher rates of eating. Clusters show moderate differences in eating behaviour which suggests that human or other activity levels and time of day influence likelihood of spotting a squirrel eating but not significantly.

A limitation is the disproportionately large number of observations recorded over the weekend compared to weekdays. This could result in the clusters reflecting the sampling pattern rather than true temporal and environmental differences since weekend and weekday conditions would likely differ.

These findings do not align with features determined important to prediction for supervised learning models. 'Hectare Conditions_Busy' and 'Shift' are not included in the top 10 features for Decision Tree but 'Shift' is the 8th most important in KNN. This is likely because clustering finds natural groupings rather than optimising prediction and pattern discovery unlike supervised learning models.

Both Clusters

The scatterplot of these clusters across the X and Y coordinates shows how the behavioural profiles, and temporal and environmental profiles are distributed across Central Park. The clusters have significant overlap indicating that these behavioural, and temporal and environmental groups coexist across the space. As clustering is an unsupervised learning method, it cannot actually predict whether a squirrel is likely to be observed eating directly. Further supervised learning methods were conducted are required to make a prediction. From Figure 3.3.5, 'Foraging' and 'Indifferent' were in the top 10 important features for decision tree while 'Running', 'Foraging' and 'Indifferent' were in the top 10 important features for KNN. These features also made up the defining features for classification

of the clusters. These findings are valuable because although clustering cannot predict eating behaviour, it revealed meaningful types of squirrels and meaningful types of environments.

There are a few limitations when performing clustering, and therefore how sound the interpretation is. Clusters are highly sensitive to feature selection. Too many dimensions (features) add noise and distract away from the main relevant features and decrease the meaningfulness of the clusters. When creating the clusters based on behaviour, there were only 10 features the squirrels were grouped against and 5 features for the temporal and environmental analysis.

5. Limitations and Improvement Opportunities

There are a few limitations with the data set used. The observations are made by human recorders which introduces subjectivity. For example, with behavioural features such as 'Kuks', 'Quaas' and 'Moans' the wrong sound could be identified or 'Running' and 'Runs From' which could have been interpreted by the observer incorrectly. Additionally, because these records are what is observed by humans, unobserved behaviours would not have been recorded thus the true behaviours, and the frequency of those behaviours are not captured in this data set. Furthermore, this data was collected 8 years ago in 2018. The data might not be relevant anymore as the squirrels' behaviour might have changed over time as environmental conditions shift and human activity changes. For example, the pandemic would likely have impacted the behaviours of the squirrels in Central Park as squirrels would have had significantly less human presence, less pollution etc. and after the pandemic was over the conditions for squirrels in Central Park would certainly have changed compared to during the pandemic but also there would be some change compared to before the pandemic as well.

Some preprocessing decisions also introduced some limitations. Removing sparse or unstructured features such as Specific Location, Litter, or Weather Notes, reduced noise and made it easier for modelling, but may have removed useful context about food sources, environment, and location within the park. This may explain why some features correlated weakly with 'Eating'. Possible improvement might be to extract temperature and weather conditions from the hectare dataset and test weather including them improves correlation strength or model performance.

6. Conclusion

This report investigated the extent to which behavioural, spatial, temporal, and environmental features can predict whether a squirrel is observed eating in Central Park. Overall, eating can be predicted to some extent, but not strongly. Correlation analysis found weak individual signals across all feature groups, with spatial and behavioural features appearing most consistently. Supervised learning supported this pattern, and the Decision Tree was more useful than KNN for detecting eating cases, drawing on both location and behaviour for prediction. Clustering further showed that temporal and environmental factors, such as PM shifts and busier hectare conditions, were associated with slightly higher eating rates, though the differences were modest. Across all methods, the available features provide some predictive signal but are not strong enough for reliable prediction on their own. These limitations came from both the modelling process, including Decision Tree overfitting and class imbalance, and the available data itself. More direct data, such as food availability, human feeding activity, visitor density, and weather conditions, would likely be needed to improve future predictive performance.

7. References

- City of New York. (2019). *2018 Central Park Squirrel Census – Squirrel Data*. NYC Open Data. <https://data.cityofnewyork.us/Environment/2018-Central-Park-Squirrel-Census-Squirrel-Data/vfnx-vebw>
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